

FOR A NEW HISTORY

SMMUN 2026

Delegates Handbook



World Wildlife Fund





World Wildlife Fund
(WWF)

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Letter by the Secretary General

Welcome to the SMMUN, a space created for each and every one of you to find your voice, your purpose, and an experience that will truly transform you. Here, every single idea matters, every perspective has value, and every bit of participation will help shape what makes this edition so unique. Beyond the debates and resolutions, this model will be an opportunity to grow, to question ourselves, to learn, and to discover everything you are truly capable of.

As Secretary General, it is an absolute privilege to welcome you to an edition that we have prepared with so much hard work, hope, and love. I deeply believe that the most meaningful experiences happen when we step out of our comfort zones, when we dare to raise our placards even when our hands are shaking, and when we understand that listening to others is also a form of leadership. At SMMUN, we aren't looking for perfection; we want to build a space where all of you can feel safe to express yourselves, make mistakes, learn, and evolve both academically and personally.

Every single detail of this edition was thought out with the intention of making each delegation feel inspired, challenged, and valued. From the committee topics to the moments you will share outside of the sessions, our purpose has always been to create an experience that leaves a lasting impression on everyone involved. Because the SMMUN doesn't just exist because of the organization behind the project, but because of the energy, the ideas, and the essence that each of you brings to this model.



Letter by the Secretary General

Because if there is one thing I have learned along this journey, it is that **nobody arrives completely ready for something like this.**

Every person who is part of a committee, a secretariat, or a dais today started out at some point doubting themselves, comparing themselves to others, and wondering if they were truly capable. **And yet... here we are.**

I have one wish for this edition, and it is this:

***May you not leave as the same person
you were when you walked in.***

I hope that when this is all over, you won't remember it just as an experience, but as something that pushed you to grow, to trust yourselves more, and to become a **better version** of who you were when you arrived.

With all my love and excitement
Marina Canto

Secretary General





Letter by the Academic Secretariat

Dear Delegate,

We are extremely happy to welcome you to this new and exciting adventure that is the SMMUN 2026.

As part of the Academic Secretariat of the model, we wholeheartedly hope that these days are wonderful for you, filled with joy, surprise, and learning and that you enjoy every moment. We want you to have fun and make the most of everything your Chair has prepared for you.

On behalf of the Academic Team, we want to tell you that the Chairs have been preparing for this moment for nearly 10 months. We have witnessed the soul, effort, and heart they have poured into this project, all with the single purpose of gifting you an experience you will always remember with great pride.

The 2026 SMMUN: For A New History marks the beginning of a new decade, a new era of changemakers. Youth with innovative ideas and different proposals, but with the same voice that must always be heard.

Always remember: “That which is not named does not exist.” Your words are the most powerful tool to bring visibility to the realities the world often prefers to ignore. By naming a problem, we give it a place in history and turn it into a shared responsibility.

Sometimes we think changing the world requires the impossible, but the truth is that great change, real change begins like this: by speaking about what truly matters, listening to others, and building solutions together. **By participating here, you are already taking that first step.**



We hope you enjoy every debate, every training session, every moment. We want you, as a delegation, to learn, to question absolutely everything, and to grow through the process. Because that is what models are: spaces for deep transformation and growth.

The delegate you aspire to become starts being built today.

Let's write this story together!

With all the love
Gali, Jaha & Day

Academic Secretariat
SMMUN 2026





Letter by the Chair

Dear Delegations,

It is with great enthusiasm and sincere appreciation that we welcome you to the **World Wildlife Fund (WWF)** committee of **SMMUN 2026**, the eleventh edition of the Southeastern Mexico Model United Nations, under the theme “**For a New History!**”.

We are truly honored to have you here and to begin this journey alongside you. Whether this is your very first Model United Nations or one of many, we want you to know that **your presence** in this committee is **deeply valued**. By choosing to be part of WWF, you are entering a space that seeks not only to encourage debate and academic growth, but also to foster collaboration and commitment toward some of the most urgent issues concerning **wildlife, biodiversity, and the future of our natural world**.

This committee seeks to be a space where ideas matter and every voice can contribute meaningfully. Through the topics we address, we hope you strengthen your skills in **debate, research, and public speaking** while developing a deeper understanding of our shared responsibility toward the environment.

As your chair, we aim to build a committee defined by respect, seriousness, and openness. We hope WWF becomes a space where first-time delegates feel **welcomed**, experienced participants feel **engaged**, and every delegation feels respected and motivated to **give their best**.

Allow us to briefly introduce the **members of the chair team**:

Diego García Flores — President

Andrea Roque Velázquez — Moderator

Susana Del Pilar Manrique Puga — Conference Officer

Mónica Helena Pat Cab — Room Officer



Each member of this chair has dedicated **time and care** to preparing this committee, and we are all genuinely excited to meet you. We have worked with the hope of creating an experience that is both enriching and memorable, and we are committed to guiding the committee process with **professionalism, warmth and responsibility**.

We would also like to thank **every delegation that joined this committee**. To those who selected WWF as their first choice, thank you for trusting in this space. To those who may be stepping into topics like these for the first time, thank you for your openness and willingness to learn. To those with previous MUN experience, thank you for bringing your preparation and perspective into this committee. Every single delegation contributes something valuable, and this committee would not be the same without **each of you**.

As we move forward, we encourage you to **prepare with dedication** and to approach this experience with **curiosity and determination**. Debate is not only about defending a position; it is also about listening carefully, thinking critically, and understanding the complexity of the issues we face.

More than anything, we hope you leave this committee having learned something that **stays with you**, both as a delegate and as a person. If WWF can be a space where you grow, contribute, and engage seriously with these issues, then it will have served its purpose.

Thank you for being part of SMMUN 2026 and for placing your trust in this committee. We are truly looking forward to meeting you and sharing this experience with you. We hope WWF becomes a space that challenges and inspires you, while leaving you with something meaningful long after the sessions are over. After all, **“The Earth is what we all have in common.”** - Wendell Berry

With respect and enthusiasm,

Diego García Flores: President, WWF Committee

Andrea Roque Velázquez: Moderator, WWF Committee

Susana Del Pilar Manrique Puga: Conference Officer, WWF Committee

Mónica Helena Pat Cab: Room Officer, WWF Committee

Kyoto the Panda: Official Committee Mascot, WWF Committee



World Wildlife Fund

(WWF)

The World Wild Fund for Nature (WWF) is one of the largest international nature conservation nonprofit organizations, founded in the Swiss Confederation more than 50 years ago on April 29 of 1961, under the idea of building a future in which humans live in harmony with nature, through sustaining the natural world. Since its foundation, WWF has invested around 10 billion dollars in more than 13,000 projects seeking to protect and restore natural habitats, stop the mass extinction of wildlife, and make the production and consumption more sustainable.

WWF emerged with a small group of enthusiasts and committed naturalists who shared one interest: nature. During the decade of the 60s, it contributed to many species and habitat conservation projects around the world. It later strengthened its commitment to international treaties related to the environment, expanding its focus to more areas related to wildlife, water and nature.

In 2023, Kirsten Schujit was named Director General of WWF International, leading and coordinating the actions of the organization. Its headquarters are located in Gland, Switzerland, and has offices in more than 80 countries around the world. WWF International is backed by some countries' own wide-ranging national WWF operations, such as WWF Mexico and WWF Spain, and is active in more than 100 countries. WWF International normally holds a reunion one time per year, however, national offices reunite several times each year and hold regular communication.

WWF is recognized as one of the most respected independent conservation organizations in the world, maintaining its independence and with all their actions determined exclusively by the organization. The organization receives funding from both public and private donors, it also works with governments and private organizations in order to achieve their goals and fulfill their projects.

Composition

The World Wild Fund for Nature is a major international non-governmental organization (NGO) with over 100 offices around the world. Its structure is composed of WWF International (located in Gland, Switzerland) and is governed by a Director General and an International Board of Trustees. WWFs governing membership does not include states or rotating delegations, but includes national offices, experts, donors and partner institutions (WWF, 2023a).



World Wildlife Fund
(WWF)

Principles of Work

WWF expresses five areas of work as the pillars of their work: diversity conservation, sustainable resource opportunities, reducing the ecological footprint of humans, decision-making based on science, and collaboration with local communities and global actors. A principle is ensuring that conservation also respects human rights and indigenous peoples (WWF, 2023b).

Agenda

WWF's agenda highlights urgent environmental issues in the following order: mitigating climate change, protecting endangered species, maintaining forests and oceans, freshwater coordination, and promoting sustainable food systems. WWF also actively participates in international negotiations, such as the UN Climate Change Conferences (WWF, 2022; Schuijt, 2023).

Faculties

- I. Promote conservation programmes for ecosystems, endangered species and biodiversity
- II. Partnership with governments and calling on world leaders in the development of sustainable public
- III. Collaborate with companies and communities to promote eco-friendly development models
- IV. Sensitize society through global campaigns that promote sustainability and environmental protection

The World Wide Fund for Nature has been working for more than 60 years on nature conservation along with communities. recognizes that sustainable management of natural resources is possible through collaboration. It began in 1961 as a small group of nature enthusiasts and is now one of the largest and most respected independent conservation organizations in the world, active in more than 100 countries and with the support of around five million people.

Outstanding achievements

Plastics: WWF encouraged citizen support for a global treaty against plastic pollution. The United Nations Environment Assembly approved the creation of a legally mandatory agreement.

Forests and Human Health: Through the Forest Vitality report, WWF demonstrated that forests not only store carbon or conserve biodiversity, but also directly influence public health, supporting nutrition and regulating diseases.

The world's largest conservation organization, urgently calls for tackling the climate and biodiversity crisis, for six decades WWF has been at the forefront of global efforts for nature, promoting innovative initiatives like the creation of emblematic protected areas.



World Wildlife Fund
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The organization has led pioneering projects such as the first eco-labels of consumption and the promotion of international agreements on wetlands, wildlife trade, biodiversity and climate change. Today, with presence in nearly 100 countries and the support of more than 35 million people. WWF is a global multicultural network that combines international leadership with local action, working to build a fairer, healthier and more sustainable world for both the planet and humanity.

Documents

As previously mentioned, the WWF is not a government organization or a political body like the UN. Therefore, it does not issue binding resolutions, treaties or law instruments. It produces a varied catalog of reports, policy briefs and campaign material for the public. Among its most influential contributions are their flagship reports, major publications that compile critical data and guide international discussions and policymaking. Most notoriously, it publishes the biennial Living Planet Report, a document describing current trends in global health and biodiversity. The main objective of this report, along with most of its other works, is to highlight threats to the natural world and influencing governments, international bodies, businesses, and the general public.





TOPICA

Measures to strengthen the regulation and traceability of wildlife trade and its biological derivatives.

Introduction to the topic

Unregulated wildlife trade represents the commercialization and markets of wild species, or their parts, in ways that infringe legal frameworks, and bypass international sustainability safeguards. This includes the trade of live animals, as well as derived products such as skins, scales, ivory, and other biological materials. Through this practice, all these can travel through complex supply chains across international borders.

A key challenge in addressing this issue is traceability, referring to the ability to verify and keep record of where a specimen came from, how it moved through the supply chain, and whether it was purchased legally. Traceability relies on multiple systems and tools, such as species identifications and standardized tracking mechanisms that allow authorities to distinguish legal trade from illegal commerce and laundering.

Traceability also implies more than identifying a species at a single checkpoint. A traceable wildlife product should provide a complete history of its origin, handling, and movement from harvest to final sale. This matters because unregulated trade does not only involve live animals or exotic pets; sectors such as fisheries, timber, and wild plants are also especially at risk. Effective traceability systems in these industries can support species conservation while also protecting the livelihoods of communities that depend on these resources.

A major reason traceability remains difficult is that wildlife products often move through multiple intermediaries and jurisdictions, which means regulation depends on consistent documentation, verification, and information-sharing across borders. In the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES), the permit system is described as the backbone of regulated international trade in listed species, but when permits and records are incomplete, inconsistent, or difficult to authenticate, illegal products can be laundered into legal channels. To address these gaps, CITES has promoted electronic permitting (eCITES), which supports electronic application and issuance, permit authentication, and standardized reporting tools that can strengthen transparency and reduce opportunities for fraud.



Despite receiving less public attention than other international crimes, the scale of this issue makes the topic incredibly urgent. Between 2015 and 2021, illegal trade across 162 countries involved roughly 4,000 species, and approximately 13 million items. This is clear evidence of regulations and enforcement challenges that affect biodiversity and the survival of threatened species.

As an organization primarily focused on conservation and protection of wildlife, WWF is at a key position regarding the international response to environmental pressures affecting species and ecosystems.

Historical Context and Background

Many of the current-day discussions regarding the aforementioned wildlife traceability back to historical tensions between conservation needs, financial systems, and law enforcement. As a key example, Kenya is often mentioned for a defining moment in 1989, when the president burned 12 tons of confiscated ivory. This act is described as a strong factor for the international ivory trade ban that happened in that same year. It also reaffirmed the country's stance against reopening ivory or rhino horn trade.

As international trade expanded, conservation concerns increasingly required cross-border regulation rather than isolated national bans. A major turning point was the creation of CITES (Convention on International Trade in Endangered Species of Wild Fauna and Flora). This document was drafted following an IUCN resolution in the 1960s, adopted in 1973, and entered into force in 1975. CITES was designed specifically to ensure that international trade in wild animals and plants does not threaten species survival, recognizing that international cooperation is essential when trade chains cross multiple jurisdictions.

CITES operationalized regulation through a system of permits and certificates required for import, export, and re-export of listed species and their parts and derivatives. This documentation framework became a central international model for traceability because it relies on standardized records as a condition for legal movement across borders. In other words, legal wildlife trade increasingly depended on paperwork and verification processes that could be checked at borders. This created a clear distinction between regulated trade and illicit flows. Over time, however, the effectiveness of this approach depended heavily on consistent implementation and verification capacity across countries.

As trade volume and complexity grew, paper-based systems also faced new pressures, including authentication challenges and slower information-sharing between authorities. In response, CITES began promoting the previously mentioned electronic permitting (eCITES), which includes electronic application and issuance of permits, permit control and authentication, and electronic reporting. Early steps included pilot efforts (notably involving Switzerland and the United Kingdom) launched around 2009, aimed at exchanging permit information electronically. This



was an important shift toward stronger verification and reduced opportunities for document fraud. This evolution is directly tied to the modern traceability agenda: traceability is not only about rules, but also about practical systems that can be validated across borders.

Over time, the demand of each of these products has also had a direct impact on international relationships and supply chains for wildlife trafficking. China, for instance, has been described as the biggest consumer of wildlife products (both legal and illegal) by multiple sources. This demand heavily increases trade in items such as ivory, rhino horns, pangolin scales, and exotic pets. More recently, in response to concerns about COVID-19's origins, China has put in place stricter regulations, including banning trade and consumption of many wild animals.

It is also essential to consider that this historical pressure from trafficking and environmental crime extends beyond “iconic” or exotic wildlife. Many regions in Africa have been prime examples of how exploitation can shift to other biological resources. Some items, such as rosewood, have been described to surpass ivory in black-market value. In countries like Madagascar, this led to the seizure of multiple products that clearly exemplify this demand.

The previously mentioned information from 2015 and 2021, also suggests that wildlife crime is not isolated to distinct regions. Instead, it has evolved over time into a widely distributed system. It operates across a wide range of countries and territories. This geographic spread also implies a level of adaptation: trafficking networks can shift routes, exploit jurisdictional gaps, and move activity across borders to reduce the risk of detection. As operations diversify across species, products, and transit points, enforcement becomes more complex. This makes coordinated, international responses even more necessary.

In the 2010s, wildlife trafficking became increasingly treated not only as a conservation problem but also as a form of transnational organized crime linked to corruption, money laundering, and document fraud. INTERPOL explicitly frames wildlife crime as exploiting the entire supply chain, from poaching and transport to processing and selling, and connects it to broader criminal activity. This shift helped drive more coordinated international efforts, including the establishment of the International Consortium on Combating Wildlife Crime (ICWC) in 2010, a partnership involving CITES, INTERPOL, UNODC, the World Bank, and the World Customs Organization to strengthen criminal justice responses. It also contributed to global political attention through the UN General Assembly's 2015 resolution (A/RES/69/314) encouraging countries to unify regulations and support the exchange of evidence. These actions further reinforced the idea that traceability and enforcement require cooperation beyond borders.

Statistical Data Supporting the Situation

Multiple countries have made public different indicators that help quantify the magnitude of illicit trade. In Indonesia, for instance, wildlife trafficking is described as being a market of wildlife valued at around \$1 billion USD annually, within the country alone.

It is also essential to emphasise that the scale of legal wildlife commerce can create conditions for laundering and enforcement overload. National reports found that 2.9 billion live animals were imported into the United States in just two decades, while the legal global wildlife trade is described as valued at \$220 billion USD. Along with China, major consumer regions identified include the U.S., EU, and Japan, demonstrating how high-volume legal demand can place constant pressure on regulation systems and complicate efforts to distinguish legal trade from illegal trafficking.

Beyond national figures, international assessments estimate that the illegal trade in fauna and flora (excluding fisheries and timber) is worth roughly \$7–23 billion USD per year, demonstrating that wildlife trafficking operates as a high-profit criminal economy. These estimates are often presented as ranges because reliable global criminal statistics are limited, but even the lower bound indicates a market large enough to sustain sophisticated trafficking networks and incentivize laundering into legal supply chains.

The scale of regulation itself also illustrates why traceability is challenging in practice. CITES keeps varying degrees of protection to over 40,900 species (thousands of animal species and tens of thousands of plant species). It is implemented by 185 Parties, meaning that traceability and documentation requirements must function across a very large number of species and jurisdictions, increasing both administrative burden and the risk of legal gaps that illegal actors can exploit.

International Measures and Efforts

All the previously presented data and background converge in one key fact: actions to tackle wildlife trade and ensure traceability rely on international, coordinated policy and enforcement. This primarily refers to having shared international standards for controlling trade and protecting biodiversity.

For instance, Kenya has been shown to be a strong advocate for strict protections for elephants and other species at CITES. Furthermore, this country has collaborated with neighbouring countries on cross-border law enforcement and sharing information to dismantle entire trafficking networks.

As a specific example of this, Kenya's port of Mombasa was once a major transit point for smuggled ivory, but increased customs inspections and international operations such as Operation Cobra have contributed to large seizures and arrests.



Other entities and countries, such as the U.S, have placed regulations that address this issue, despite not being in direct collaboration with other countries. A clear example of this, is this country's Lacey Act, which restricts imports of illegally sourced wildlife or wood and supports regional anti-trafficking efforts.

The U.S., however, has also been shown to direct several international operations; agencies including the U.S. Fish & Wildlife Service and Homeland Security routinely make large seizures and maintain special representatives working abroad. It is also noted that in 2016 the U.S. essentially banned domestic ivory sales, and that American tech companies cooperate in efforts to shut down online wildlife markets, reflecting the increasing importance of digital enforcement strategies.

Another example of coordinated policy at scale is the European Union, which has treated wildlife trafficking as a cross-border priority requiring action among member states. The EU adopted an Action Plan against Wildlife Trafficking in 2016 and later introduced a revised action plan intended to strengthen ambition and serve as a blueprint for cooperation, within the EU and internationally, to reduce illegal wildlife trade. This kind of regional framework supports traceability goals by encouraging more consistent enforcement standards, stronger cooperation between authorities, and a shared approach to preventing illegal products from entering legal markets.

Current Situation

Current wildlife trade dynamics are increasingly shaped by the interaction between high-demand consumer markets and adaptable trafficking networks. Multiple countries, such as the United States, take the role as both major consumers and major enforcers. These countries remain as a large legal market for wildlife and wildlife products, while also serving as a destination for illegally sourced items such as ivory, reptile skins, and illegally harvested timber. This dual role complicates regulation, because high-volume legal flows can create opportunities for laundering and overwhelm monitoring systems.

At the same time, enforcement is being pushed to adapt to evolving methods and channels. The documents highlight efforts to address online trade, including Indonesia's 2024 focus on targeting online wildlife sales and transnational smuggling rings, and U.S.-linked cooperation by tech companies in shutting down online wildlife markets. These examples reinforce that contemporary wildlife crime is not only international, but increasingly facilitated by digital marketplaces and decentralized supply chains, making traceability standards and coordinated enforcement more essential, especially in digital environments.

Recent international assessments emphasize that wildlife trafficking continues worldwide despite years of enforcement and policy efforts. UNODC's World Wildlife Crime Report 2024 notes that seizures alone are not enough to understand or end the problem, and that traffickers can be quick to adapt their methods in response to pressure. This reality helps explain why traceability remains central: as networks evolve, enforcement must rely on systems that can follow products across multiple steps and jurisdictions, rather than only intercepting them at isolated points.

Relevance and Urgency

Wildlife trade regulation and traceability are urgent because unregulated extraction and commerce represent a direct threat not only to individual species, but also to ecosystem stability and human well-being. Unregulated wildlife trade has been repeatedly shown to be able to disrupt ecological balance and undermine the natural systems that support climate stability. It is also described as being driven by short-term financial gains that can degrade biodiversity and harm the communities that rely on wildlife for sustainable livelihoods.

In practice, the consequences of weak traceability are not limited to traditional wildlife products; high-value biological resources can become drivers of persistent exploitation when governance is fragile. Madagascar is used as an example of how poverty and weak enforcement can intensify illegal extraction: rare species and resources can become a concrete source of income in a very poor country, reinforcing the cycle of biodiversity loss. Initiatives such as community forestry and ecotourism are being pursued as more sustainable alternatives, highlighting that reducing illegal trade pressure often requires economic pathways that do not depend on trafficking.

Ultimately, strengthening regulation and traceability is not simply a technical upgrade, it is a necessary condition for making conservation enforceable in real markets. Without transparent supply chains and credible verification, illegal products can continue to move through loopholes, undermining both biodiversity protection and the long-term economic security of communities that depend on legal and sustainable use.

Pillars

Stronger Laws and Regulations

- Updating and aligning regulations so wildlife trade is clearly defined, loopholes are reduced, and legal trade is easier to verify.

Traceability and Transparency

- Ensuring products can be tracked from origin to final sale through reliable records and verification, so illegal goods can't be laundered into legal markets.

Enforcement and Penalties

- Improving inspections, investigations, coordination between authorities, and consequences for trafficking to make enforcement effective in practice.

Online Trade and Technology

- Addressing digital marketplaces and using tools like e-permitting, authentication, and identification support to strengthen traceability, especially online.

Sustainable Trade and Communities

- Protecting livelihoods that depend on legal, sustainable trade and reducing incentives that drive illegal extraction and trafficking.



TOPIC A'S GUIDE QUESTIONS

- ★ How effective are national and international legal frameworks that regulate wildlife trade in practice?
- ★ How can technology strengthen traceability systems?
- ★ What challenges do Member States face in enforcing wildlife trade regulations?
- ★ What responsibilities do consumer countries have in reducing demand for illegal wildlife products?
- ★ How can international cooperation and information sharing be improved to oppose transnational wildlife trade?
- ★ How do differences in national legislation create loopholes in international wildlife trade regulation?
- ★ How can consumer awareness and demand reduction strategies support stronger regulation of wildlife trade?
- ★ How does wildlife trade intersect with other global challenges such as climate change and public health?



TOPIC A'S GLOSSARY



Traceability:

refers to the ability to track the origin, movement and final destination of wildlife specimens and their derivatives throughout the supply chain.



Biological derivatives:

substances derived from living organisms or natural sources.



Wildlife forensics:

it is concerned with providing scientific evidence to inform investigations into crimes against wildlife, focusing on determining the identity of unlawfully traded wildlife products. It addresses questions related to the species, geographic origin, relatedness, among others.



Transit country:

a country in which wildlife products pass in route from source to destination.



Non-Detriment Finding (NDF):

a scientific assessment ensuring that wildlife trade does not threaten species survival.



Zoonotic disease:

infectious illnesses that spread between animals and humans, often linked to unregulated wildlife trade due to the lack of preventive measures.



Chain of custody:

a process that tracks the movement of evidence by documenting each person who handled the evidence, the date it was collected or transferred and the purpose for the transfer.



Species listing:

official process in which a plant or animal species is added to a government-maintained list of endangered or threatened species. Listed species receive legal protections under regulations.



RECOMMENDED READINGS

1. **WWF Wildlife Trade and Zoonotic Disease Risk Studies:**
<https://www.worldwildlife.org/our-work/wildlife/reducing-zoonotic-disease-risk-from-wildlife-trade/>
2. **Convention on International Trade in Endangered Species (CITES) Permanent Committee update 2025: :**
<https://cites.org/sites/default/files/eng/com/sc/79/E-SC79-ES.pdf>
3. **CITES press release on sustainable, legal and traceable wildlife trade:**
<https://cites.org/eng/news/cites-at-cbd-cop16-2024>
4. **UNODC World Wildlife Crime Report:**
https://www.unodc.org/documents/data-and-analysis/wildlife/2024/Wildlife2024_Final.pdf
5. **World Customs Organization (WCO) Illicit Trade Report:**
https://www.wcoomd.org/-/media/wco/public/global/pdf/to-pics/enforcement-and-compliance/activities-and-programmes/illicit-trade-report/itr_2024_en.pdf?db=web



TOPIC B

Actions to Safeguard Genetic Diversity and Ensure Equitable Access to Genetic Resources, Addressing the Accelerating Loss of Ecological Variety

Introduction to the topic

Genetic diversity is the foundation of life on Earth, however, it is disappearing at an alarming rate as climate change, deforestation, industrialization and overexploitation intensify. It is estimated that there has been a 73% decline in the average size of monitored wildlife populations in just 50 years. Besides, over one million species are currently at risk of extinction. When a species disappears, its unique genetic information is lost forever, with no chances of recovering it and weakening entire ecosystems around the world.

At the same time, genetic resources, including seeds, plants and animal genomes, are mainly controlled by a few countries and corporations, creating deep inequalities in access and benefits. Many countries with developing economies, where the majority of the world's biodiversity is centered, lack the technology, funding or legal protection required to preserve species' genetic heritage and to fairly benefit from its use.

This topic addresses two urgent challenges: protecting genetic diversity from rapid ecological loss and ensuring equitable access to genetic resources from those species for all nations. Safeguarding genetic variety is essential not only for environmental stability, but also for global food security, public health, sustainable development and, with an ethical biotechnologic and scientific approach, a future restoration of those genetically preserved species. All of these while regulating equitable access to genetic resources of all species for every country and preventing biopiracy.

Without international cooperations, the loss of genetic diversity will become irreversible, threatening both natural ecosystems and human survival. As the World Health Organization mentioned "biodiversity is essential for our survival". For this, it must be addressed with a cross-sectoral, political, equitable and ethical perspective to safeguard biodiversity for future generations.

Historical Context and Background

The relationship between human societies and genetic diversity dates back more than 10,000 years, around 8000 BCE, when early agricultural communities began the domestication of plants and animals. Indigenous peoples and traditional farming communities played a crucial role in conserving seeds, animal breeds, and ecological knowledge, establishing the foundations of food security and ecosystem resilience long before the emergence of modern states.

This long-standing balance began to shift significantly during the Industrial Revolution (1760–1840). Fast industrialization, population growth, and urban expansion intensified the exploitation of natural resources. Large-scale land conversion, deforestation, and mechanized agricultural practices disrupted ecosystems and reduced habitat availability for many species. Although productivity increased, these changes marked the beginning of a steady decline in genetic diversity, as standardized production methods replaced diverse local systems.

The trend accelerated further during the mid-20th century, particularly between the 1940s and 1970s, with the advent of the Green Revolution. High-yield crop varieties, combined with chemical fertilizers, pesticides, and irrigation, transformed global food production and helped prevent widespread famine. However, this progress came at a significant cost: monoculture farming systems expanded rapidly, displacing thousands of traditional crop varieties and contributing to severe genetic erosion. As agricultural systems became more genetically uniform, their vulnerability to pests, diseases, and environmental stress increased.

By the 1970s, growing scientific evidence and public concern brought environmental degradation and biodiversity loss to the forefront of international debate. The 1972 United Nations Conference on the Human Environment in Stockholm marked a turning point, as it formally linked environmental protection with economic and social development. During this period, researchers increasingly emphasized that genetic diversity was essential not only for conservation, but also for long-term sustainability and human survival.

A major milestone was reached in 1992, when the international community adopted the Convention on Biological Diversity (CBD) at the Earth Summit in Rio de Janeiro. The CBD fundamentally reshaped global biodiversity governance by recognizing the conservation of biological diversity as a shared responsibility and by affirming national sovereignty over genetic resources. It introduced key principles such as prior informed consent and fair and equitable benefit-sharing, addressing long-standing concerns over the exploitation of biological resources from biodiversity-rich countries.

Building on this framework, the International Treaty on Plant Genetic Resources for Food and Agriculture, adopted in 2001, acknowledged the historical contributions of farmers to the development and conservation of crop diversity. The treaty established a multilateral system to facilitate access to plant genetic resources

essential for global food security while promoting benefit-sharing and cooperation among states.

The adoption of the Nagoya Protocol in 2010, which entered into force in 2014, further strengthened the access and benefit-sharing regime under the CBD. It sought to prevent biopiracy and ensure that benefits derived from genetic resources and associated traditional knowledge were shared fairly with provider countries, as well as Indigenous peoples and local communities who have long acted as stewards of biodiversity.

Since 2015, with the adoption of the 2030 Agenda for Sustainable Development and the Paris Agreement on climate change, genetic diversity has been increasingly recognized as a critical component of climate adaptation and ecosystem resilience. In the 2020s, rapid advances in biotechnology, genome sequencing, and the use of digital sequence information have introduced new ethical and legal challenges, as genetic data can now be accessed and utilized without direct physical access to biological material.

Overall, the historical evolution of genetic diversity protection illustrates a gradual transition from unregulated exploitation toward structured international cooperation. This progression underscores the growing recognition that safeguarding genetic diversity and ensuring equitable access to genetic resources are essential for addressing the accelerating loss of ecological variety and for promoting long-term global sustainability.

Statistical Data Supporting the Situation

Genetic variability the variability of genes within species is a basic component of biodiversity that can help populations successfully adapt to environmental changes, epidemiological emergencies, or extreme weather events. However, genetic diversity is declining across the globe, posing an urgent challenge to natural systems and human food security.

Loss of genetic diversity on a global scale

A global analysis published in the journal *Nature* has analyzed more than 30 years of data on the genetics of 628 species of animals, plants, fungi, and other organisms, and has found that approximately two out of three populations have lost genetic diversity during this period.

This loss of diversity can be observed not only in endangered species, but also in those that are not currently considered as such, indicating the existence of a widespread phenomenon among all wildlife.

Agricultural genetic resources in danger

Regarding advances in genetic resources for food and agriculture, the Food and Agriculture Organization of the United Nations (FAO) reports that, in December 2024,

only ex situ genetic material (i.e., in germplasm banks) was preserved for:

- 365 of 7,809 breeds and varieties of domestic animals and livestock ($\approx 5\%$ of the total).
- 194 of 1,126 transboundary breeds and varieties ($\approx 17\%$ of the total).

This indicates a very low availability of genetic resources conserved over time, which limits recovery in the face of future environmental pressures.

Importance of biological diversity for human security

More than 50% of modern drug therapy comes from natural sources, and about 75% of food crops depend on pollinators (this is an example of associationism: pollinators are associated with a certain level of genetic diversity in plants) for production.

Nearly 35% of the planet's wetlands have been lost since 1970, ecosystems that are key to maintaining genetic resources and environmental services such as freshwater and habitat for biological diversity.

Loss of local animal breeds

Of the more than 7,000 local livestock breeds that exist in the world, approximately 29% are in danger of extinction, indicating a significant erosion of the animal genetic biodiversity that sustains global agricultural and food systems.

International Measures and Efforts

The international community has become aware of the fact that the accelerated loss of genetic and biological diversity jeopardizes ecological stability, food security, and sustainable development. It has therefore established various regulatory frameworks and multilateral cooperation mechanisms for the conservation of genetic resources and to guarantee equitable access to the benefits derived from their use.

The main instrument in this area is also the Convention on Biological Diversity, which was adopted in 1992 within the framework of the United Nations and establishes three fundamental objectives: the conservation of biological diversity, the sustainable use of its components, and the fair and equitable sharing of the benefits arising from the use of genetic resources. Through the CBD, the States Parties recognize the national sovereignty of their genetic resources and, at the same time, recognize the need to cooperate for the development, protection, and responsible management of biological resources.

In turn, the Nagoya Protocol, in force since 2014, is a binding legal framework that regulates access to genetic resources and the equitable sharing of benefits derived from their use. It aims to prevent biopiracy while strengthening the rights of indigenous and local communities by recognizing their historical role in the conservation of genetic diversity and associated traditional knowledge.



In terms of food security, the FAO has promoted the Treaty, which aims to ensure the conservation and exchange of essential plant genetic resources. This treaty promotes a multilateral system of access to plant genetic resources that facilitates international cooperation and, at the same time, strengthens the resilience of agricultural systems to the effects of climate change.

In addition, the FAO has implemented mechanisms for the protection of animal genetic diversity, such as the Domestic Animal Diversity Information System (DAD-IS), which provides information on the status of livestock breeds worldwide and assists decision-making in the field of in situ and ex situ conservation.

Finally, the Kunming-Montreal Global Biodiversity Framework, agreed upon in 2022, symbolizes a renewed commitment to halting and reversing biodiversity loss by 2030. This framework contains clear objectives, such as the conservation of the genetic diversity of both wild and domesticated species and highlights the role of international cooperation as the driving force behind the achievement of these objectives. WWF, for example, has adopted these goals within conservation projects, scientific research, and work with local communities in order to strengthen efforts for global biodiversity governance.

Current Situation

2020s. Scientific assessments conducted under the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES) have confirmed that genetic diversity is declining across species at an unprecedented rate, undermining ecosystem resilience, food security, and the capacity of societies to adapt to climate change.

At the multilateral level, the Convention on Biological Diversity (CBD) remains the central legal framework guiding global action. A major milestone was reached in December 2022 with the adoption of the Kunming–Montreal Global Biodiversity Framework (GBF) at the 15th Conference of the Parties (COP15) in Montreal. This framework set ambitious global targets to be achieved by 2030, including the conservation of genetic diversity of wild and domesticated species and the safeguarding of Indigenous and local knowledge associated with genetic resources. The GBF explicitly recognizes genetic diversity as essential for ecosystem functionality and sustainable development, marking a shift toward a more integrated biodiversity governance approach.

In parallel, the implementation of the Nagoya Protocol continues to be a central issue in international negotiations. Since its entry into force in 2014, Parties have met regularly through the Meetings of the Parties to the Nagoya Protocol (COP-MOP) to address challenges related to access and benefit-sharing (ABS). Recent discussions, particularly between 2020 and 2024, have focused on improving compliance mechanisms and addressing gaps that disadvantage biodiversity-rich developing countries and Indigenous peoples.



One of the most complex and ongoing debates concerns Digital Sequence Information (DSI). As genomic technologies advanced rapidly during the 2010s and 2020s, genetic data increasingly became accessible through digital databases without the physical transfer of biological material. This issue gained prominence during CBD COP15 (2022), where Parties agreed to continue negotiations on a multilateral benefit-sharing mechanism for DSI. Further technical and political discussions have since taken place in intersessional working groups and at subsidiary body meetings under the CBD, highlighting the evolving nature of genetic resource governance.

The Food and Agriculture Organization of the United Nations (FAO) plays a critical role in addressing genetic diversity in food and agriculture. Through the Commission on Genetic Resources for Food and Agriculture (CGRFA), established in 1983, FAO coordinates global efforts to conserve plant, animal, forest, and aquatic genetic resources. Periodic global assessments and action plans—updated throughout the 2010s and 2020s have documented severe genetic erosion in crops and livestock, while promoting on-farm conservation and sustainable use strategies.

International conservation organizations, particularly the World Wide Fund for Nature (WWF) and the International Union for Conservation of Nature (IUCN), have contributed significantly to current efforts. WWF has consistently highlighted genetic diversity loss in its Living Planet Reports, published every two years, with recent editions emphasizing the link between genetic erosion, ecosystem collapse, and climate change. Meanwhile, IUCN integrates genetic considerations into its Red List assessments and supports conservation policies that recognize genetic diversity as a core component of species survival.

At the regional and national levels, implementation remains uneven. While some states have strengthened ABS legislation and expanded protected areas following COP15, others continue to face capacity, funding, and governance constraints. International financial mechanisms, including the Global Environment Facility (GEF), have increased funding allocations for biodiversity and genetic conservation projects during the 2020s, yet demand continues to outpace available resources. In addition, Indigenous peoples and local communities have gained greater recognition within international forums. Since 2018, the CBD's Working Group on Article 8 has increasingly emphasized the role of traditional knowledge in genetic conservation. This trend was reinforced at COP15, where Parties reaffirmed the importance of full and effective participation of Indigenous peoples in biodiversity governance and benefit-sharing arrangements.

In conclusion, the current situation reflects both progress and persistent challenges. International organizations, treaties, and conferences have established a robust institutional architecture to address genetic diversity loss, particularly since 2022. However, accelerating ecological pressures, unresolved issues such as DSI, and persistent inequalities in access and benefit-sharing continue to limit the effectiveness of global action. Addressing these challenges will require sustained political commitment, enhanced international cooperation, and adequate financing to translate global agreements into meaningful outcomes on the ground.

Relevance and Urgency

The rapid decline in genetic diversity and biodiversity represents an urgent and priority challenge for the international community, as it jeopardizes the stability of ecosystems and the planet's ability to maintain conditions for human life.

Genetic diversity enables species to adapt to environmental changes, resist disease, and fulfill their substantive ecological functions; its erosion reduces the resilience of ecosystems to climate change, environmental degradation, and extreme events, putting ecosystems at risk of irreversible collapse. The urgency of the issue is also reinforced by its consequences for food security and the livelihoods of millions of people.

The genetic homogenization of crops and the loss of local breeds increase the vulnerability of global food systems to pests, droughts, and climate crises, which disproportionately affect developing countries and rural communities in particular. Furthermore, the loss of genetic resources limits future opportunities for scientific research, drug development, and technological innovation, deepening inequalities between states and regions.

From a global perspective, failure to act on the erosion of genetic diversity jeopardizes the achievement of the Sustainable Development Goals, particularly zero hunger, climate action, and life on land and in water ecosystems. Without coordinated action, conflicts over natural resources would increase, economic insecurity would rise, and, at the same time, the risk of weakening environmental governance on a global scale would increase. The protection of genetic diversity must be considered a shared responsibility and a prerequisite for ensuring ecological stability, human well-being, and long-term sustainable development.

Pillars

Conservation of Genetic Diversity

This pillar is focused on preventing the extinction of species and the erosion of genetic diversity within populations around the world. The protection of habitats needs to be addressed in order to create a discussion around other subtopics.

Equitable Access and Sharing

Fair access to genetic resources for ethical uses is highly relevant to avoid monopolies or biopiracy. Each Member State should make sure of the implementation of treaties, protocols or laws, such as the Nagoya Protocol, to ensure that every community benefits from the use of biodiversity, guaranteeing transparency, consent and no possibility of patents.

Sustainable Use of Genetic Resources

It is crucial to prevent the exploitation of species and their genetic information for either commercial or scientific purposes. The regulation of genetic resources through the balancing between economic funding with long-term ecosystem resilience leads to the avoidance of a possible depletion of them.

Restoration and Recovery of Species

The rewilding and ecosystem restoration for current endangered species could be addressed through genetic rescue, breeding programs and the use of advanced science, such as genetic mapping, assisted reproduction and seed vaults.

International Cooperation and Capacity Building

Scientific collaboration and data sharing between all Member States, creating global databases with equitable access and the strengthening of research networks is the only way of preserving all species around the globe.





TOPIC B'S GUIDE QUESTIONS



Why is genetic diversity essential for ecosystem resilience and long-term species survival?



How effective are existing international agreements in protecting genetic resources?



How can Member States prevent biopiracy while still encouraging scientific research and innovation?



What strategies can be implemented to preserve genetic diversity in situ and ex situ?



What obligations do biodiversity-rich countries and technology-rich countries have toward each other regarding genetic resources?



- ★ In what ways does unequal access to biotechnology deepen global inequalities related to genetic resources?
- ★ What role do gene banks and seed vaults play in preventing irreversible genetic loss, and what are their limitations?
- ★ What ethical boundaries should predominate genetic modification and synthetic biology in biodiversity conservation?
- ★ What accountability mechanisms should exist when genetic resources are misused or benefits are unfairly distributed?



TOPIC B'S GLOSSARY



Genetic diversity:

refers to the ability to track the origin, movement and final destination of wildlife specimens and their derivatives throughout the supply chain.



Genetic resources:

they are defined in the CBD as genetic material of plant, animal, microbial or other origin containing functional units of hereditary that has actual or potential value, and do not include human genetic resources.



Biopiracy

concept referring to the unethical or unlawful appropriation or commercial exploitation of biological materials that are native to a particular country or territory without providing fair financial compensation to the people or governments of that territory.



In situ conservation:

defined as the conservation of organisms or ecosystems in their natural surroundings, maintaining species diversity at the gene level and supporting evolutionary adaptations, often within protected areas.



Ex situ conservation:

refers to the maintenance of genetic variation away from its original locations. It is the most convenient and widely used method of genetic conservation.



Access and benefit sharing (ABS)

according to the CBD, it refers to the way in which genetic resources may be accessed, and how the benefits that result from their use are shared between the users of the resources and the providers of those resources.



Bonn Guidelines:

voluntary guidelines intended to assist governments in the adoption of measures to govern access and benefit-sharing in their countries.



Mutually agreed terms (MAT):

agreements reached between the providers of genetic resources and users on the conditions of access and use of the resources. The benefits are shared between both parties.



Competent National Authorities (CNAs):

bodies established by governments, responsible for granting access to users of their genetic resources, and representing providers on different levels.



Prior informed consent (PIC):

the permission given by the CNA of a country to an individual or institution seeking to obtain access to genetic resources, aligning with appropriate legal and institutional framework.



RECOMMENDED READINGS

1. **Convention on Biological Diversity (CBD):**
<https://www.cbd.int/doc/legal/cbd-en.pdf>
2. **Nagoya Protocol:**
<https://www.cbd.int/abs/doc/protocol/nagoya-protocol-en.pdf>
3. **Le Monde article on Genetic Resources at COP16:**
https://www.lemonde.fr/en/environment/article/2024/10/22/biodiversity-sharing-the-benefits-of-genetic-resources-a-sensitive-issue-at-cop16_6730140_114.html#
4. **Bonn Guidelines from CBD:**
<https://www.cbd.int/doc/publications/cbd-bonn-gdls-en.pdf>



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